Showkot Hossain

Ph.D. Student, Computer Science and Engineering, University of Notre Dame

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Research Interests

Security and privacy, distributed systems, confidential computing

Education

Aug 2024 – Present — **Ph.D. in Computer Science and Engineering** — *Notre Dame, IN, USA*
University of Notre Dame
Advisor: Dr. Taeho Jung · GPA: 3.91/4.0

Jan 2016 – Jan 2020 — **B.Sc. in Computer Science and Engineering** — *Dhaka, Bangladesh*
University of Dhaka
Thesis: Network Log Management System using Blockchain
Advisors: Dr. Mosarrat Jahan and Dr. Upama Kabir

Publications

Conference Papers

[C6] **Decentralized AI Governance Must Decouple Policy Processing from Capability Enforcement**
Hasan Kassem, Christoforos Anagnostopoulos, Alejandro Aristizabal, Spyridon Bakas, Orion GB Banks, Omar Benjelloun, Tian Cai, Sergen Cansiz, Davide Chicco, Patrick Foley, **Showkot Hossain**, Taeho Jung, Peter Kairouz, Yanchen Liu, Marco Lorenzi, Peter Mattson, Ann K Novakowski, Michael O'Connor, Holger R Roth, Adrish Sannyasi, Charalampos Savvaidis, Micah Sheller, David Solooki, Dimitris Stripelis, Renato Umeton, Marc Vesin, Ray Wang, Wenbin Zhang, Mic Bowman, Alexandros Karargyris
40th Annual Conference on Neural Information Processing Systems (NeurIPS 2026), Position Paper Track.
Acceptance rate: 22.9%

[C5] **Cost-Effective Logging for Decentralized Massive Data Sharing**
Avery Hughes, **Showkot Hossain**, Yanchen Liu, Changhao Chenli
9th IEEE International Conference on Blockchain (IEEE Blockchain 2026)

[C4] **Web3IoT: A Decentralized Database Engine for Secure and Scalable IoT Data Management**
Omkar Rokdoba Doijad, Chandan Monabolu Narayana, **Showkot Hossain**, Jiannong Cao, Taeho Jung, Haijian Sun, WenZhan Song
IEEE International Conference on E-health Networking, Application & Services (IEEE HealthCom 2026)

[C3] **MtDB: A Decentralized Multi-Tenant Database for Secure Data Sharing**
Showkot Hossain, Wenyi Tang, Changhao Chenli, Haijian Sun, WenZhan Song, Seokki Lee, Mic Bowman, Taeho Jung
8th IEEE International Conference on Blockchain (IEEE Blockchain 2025), pp. 79–86

[C2] **SecLog: Atomic-Swap Based Secure Logging for Decentralized Data Sharing**
Avery Hughes, **Showkot Hossain**, Wenyi Tang, Taeho Jung, Changhao Chenli
International Conference on Metaverse (METAVERSE 2025), Springer, pp. 115–124

[C1] **Towards Blockchain-Driven Network Log Management System**
MH Rakib, **Showkot Hossain**, Mosarrat Jahan, Upama Kabir
IEEE 8th International Conference on Smart City and Informatization (iSCI 2020)

Journal Articles

[J2] **Consentful-by-Design: A Perspective on Safeguarding Data Ownership from Generative AI Leveraging Lessons from the Healthcare Domain**
Akis Linardos, Theresa Willem, Alena Buyx, **Showkot Hossain**, Taeho Jung, Dimitrios Makris, Spyridon Bakas
AI & SOCIETY, Springer, pp. 1–13, 2026

[J1] **A Blockchain-Enabled Scalable Network Log Management System**
MH Rakib, **Showkot Hossain**, Mosarrat Jahan, Upama Kabir
Journal of Computer Science, 18(6), pp. 496–508, 2022

Research Experience

Aug 2024 – Present
Graduate Research Assistant *Notre Dame, IN, USA*
Data Security and Privacy Lab (DSP-Lab), University of Notre Dame
Advisor: Dr. Taeho Jung
- Research and develop Web3DB, a decentralized platform for secure data sharing and management, and Web3FS, its decentralized file system, as part of a collaborative team; published several conference papers.

Jan 2020 – Dec 2020
Research Assistant *Dhaka, Bangladesh*
Cyber Security Lab, University of Dhaka
Advisors: Dr. Mosarrat Jahan and Dr. Upama Kabir
- Built a prototype system for secure provenance tracking and real-time monitoring using PKI and blockchain; published one peer-reviewed journal article and one conference paper.

Industry Experience

Nov 2023 – Jul 2024
Software Engineer, Level 3 *Dhaka, Bangladesh*
Itransition Software Bangladesh Limited
- Collaborated with global clients to design and implement high-performance APIs for Product Madness, a leading mobile game studio with 5M+ monthly active users, consistently meeting release deadlines.
- Led a team of 5 developers building and maintaining features with Spring Boot (Java), React, and PostgreSQL in an Agile process (JIRA, Git, JUnit, Mockito).

Mar 2023 – Nov 2023
Software Engineer, Level 2 *Dhaka, Bangladesh*
Kona Software Lab Limited
- Implemented the authentication and authorization system for microservices serving 13M+ KonaCard users.
- Redesigned the database architecture and wrote a migration script, reducing query latency by 25%.

Oct 2020 – Mar 2023
Software Engineer, Level 1 *Dhaka, Bangladesh*
Dohatec New Media
- Researched and developed a microservices-based backend for ePMIS, the Government of Bangladesh's Electronic Project Management Information System, built for high performance and scalability.
- Designed and developed high-performance RESTful APIs with Spring (Java), AngularJS, and SQL Server in cross-functional Agile teams.

Jun 2020 – Aug 2020
Blockchain Developer (Intern) *Dhaka, Bangladesh*
Technohaven Company Limited
- Studied blockchain architecture, consensus algorithms, digital identity, and blockchain applications.
- Developed a prototype supply-chain traceability system using Hyperledger Fabric [code].

Honors, Awards & Grants

2025 **Travel Grant**, University of Notre Dame. $1,532 to present at IEEE Blockchain 2025 (Zhengzhou, China).

2024 **Travel Grant**, University of Notre Dame. $1,000 to attend ACM CCS 2024 (Salt Lake City, UT, USA).

2024 – 2029 **Research Assistantship**, University of Notre Dame. Full tuition, health insurance, and stipend.

2020 – 2021 **ICT Innovation Fund**, ICT Division, Government of Bangladesh. $5,000 grant for innovative project development.

2020 **Award of Merit**, International Blockchain Olympiad (team of 4).

2020 **9th Place**, Blockchain Olympiad Bangladesh (team of 4).

2011 – 2020 **Government Education Board Scholarships**, Bangladesh. Awarded three times for national examination results: top 1% in the Higher Secondary Certificate (2015), top 2% in the Secondary School Certificate (2013), and top 1% in the Junior School Certificate (2010).

Teaching

Fall 2025 **Teaching Assistant**, CSE 20110: Discrete Mathematics, University of Notre Dame

Mentoring

Undergraduate researchers, co-supervised with Dr. Taeho Jung.
*University of Notre Dame · †Indiana Institute of Technology

Spring 2026	Will Armsworthy*, Paola Lopez*, Aaron Wang*
Fall 2025	Will Armsworthy*, Maria C. Leal*, Avery Hughes†
Spring 2025	Ryan Nahm*, Aaron Wang*, Avery Hughes†
Fall 2024	Gustavo Aniceto*, Jonathan D'Alonzo*

Professional Service

2026	Reviewer, ACM Distributed Ledger Technologies (DLT)
2026	Reviewer, IEEE Transactions on Dependable and Secure Computing (TDSC)
2026	Reviewer, IEEE Network
2025 – 2026	Reviewer, IEEE International Conference on Blockchain and Cryptocurrency (ICBC)
2025	Reviewer, IEEE International Conference on Blockchain (Blockchain)

Leadership & Outreach

2025 – 2026	**Sports & Cultural Secretary**, Bangladeshi Students Association of Notre Dame (BDSA-ND)
2017 – 2024	**Co-founder**, Jiboner Alo. Education for orphaned and underprivileged children.
2016 – 2021	**Tutor**, private tutoring for high school and undergraduate students.
2016 – 2020	**Executive Member**, Badhan (Dr. Muhammad Shahidullah Hall, University of Dhaka). Organized blood donation drives and educational workshops.
2019	**Student Volunteer**, International Conference on Innovation and Engineering (ICIET)
2018	**Mentor**, SCB-PA Inter School and College Programming Contest (ISCPC)

Technical Skills

Languages	Python, Java, C/C++, JavaScript, Solidity
Systems & Web	Spring Boot, React, Node.js, FastAPI, Parse, JPA, Hibernate, RabbitMQ, Kafka, ELK Stack, Gradle, JUnit, Mockito
Databases	PostgreSQL, Microsoft SQL Server, MongoDB, Redis
ML	Pandas, NumPy, PyTorch, TensorFlow
Blockchain	Smart contracts (Solidity), Hyperledger Fabric
Tools	Trusted Execution Environments (TEE), Docker, Jenkins, Nexus, Postman, Swagger, JMeter, Jira, Git

Selected Projects

PostStream	A micro-blogging application inspired by Twitter, built with React and Parse Server, with real-time content discovery, social interactions, and dual-platform deployment [code].
Emotion	A BERT-based multi-class classifier for 13 tweet emotions, improving accuracy on an imbalanced dataset through synonym augmentation, ChatGPT-generated synthetic data, and class filtering [code].
Spam Detector	A DistilBERT-based spam detector on the Enron dataset reaching 99.36% accuracy with balanced precision and recall, outperforming traditional ML and deep-learning baselines [code].

References

Dr. Taeho Jung
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Department of Computer Science and Engineering
University of Notre Dame, IN, USA
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Ph.D. advisor

Dr. WenZhan Song
Professor
School of Electrical and Computer Engineering
University of Georgia, Athens, GA, USA
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Research collaborator